

```
config-3.3.0-1800-linaro-lt-vexpress-a9
initrd.img-3.3.0-1800-linaro-lt-vexpress-a9
System.map-3.3.0-1800-linaro-lt-vexpress-a9
vmlinuz-3.3.0-1800-linaro-lt-vexpress-a9
```

Copy the original VExpress images, `vmlinuz.*` and `initrd.*`, somewhere where you plan to keep your VM.

Now run the image using QEMU:

```
$ qemu-system-arm -kernel vmlinuz-3.3.0-1800-linaro-lt-
vexpress-a9 -M vexpress-a9 -cpu cortex-a9 -serial stdio -m
1024 -initrd initrd.img-3.3.0-1800-linaro-lt-vexpress-a9
-append 'root=/dev/mmcblk0p2 rw mem=1024M raid=noautodetect
console=tty1 rootwait vmalloc=256MB devtmpfs.mount=0' -redir
tcp:5022::22 -sd vexpress.img
```

This will open the window where your ARM based VM is booting, like any other normal operating system.

- **-redir tcp:2200::22** redirects TCP traffic on the host port 2200 to the guest machine's (QEMU) Port 22. This will allow us to SSH into the machine later by connecting to localhost on Port 2200.
- **-m 512** specifies that we want 512MB of RAM. You can adjust this, but make sure you also change it in the '-append' string.
- **-drive file=vexpress.img,if=sd,cache=writeback** attaches

the images as an SD card. According to people at Linaro, this method provides faster I/O compared to the traditional `-sd` option.

After starting the machine, you should end up with a Linux shell in the QEMU window after some startup time.

Installing the required packages inside the VM

You now need to configure the VM for your general use. First install some common packages, as follows:

```
$ apt-get update
$ apt-get install binutils gcc gdb build-essential vim
openssh-server
```

You can also connect to this VM using `ssh` in future. For that, start the `ssh` service and set the root password using the `passwd` command.

Our complete virtual environment is now ready. You can use it in the same way you use any other Linux distro for development or general purposes. **END** 🐧

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